

# Kasra Ferdowsi

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## Education

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| <b>UC San Diego</b>                                                                                                                               | 2019 - 2024 |
| <ul style="list-style-type: none"><li>- PhD in Computer Science and Engineering</li><li>- Advised by Nadia Polikarpova and Sorin Lerner</li></ul> |             |
| <b>UC Berkeley</b>                                                                                                                                | 2017 - 2019 |
| <ul style="list-style-type: none"><li>- B.S. in Electrical Engineering and Computer Science</li><li>- Graduated with Highest Honors</li></ul>     |             |
| <b>Diablo Valley College</b>                                                                                                                      | 2014 - 2017 |
| <ul style="list-style-type: none"><li>- A.S. for Transfer in Mathematics</li><li>- A.S. for Transfer in Physics</li></ul>                         |             |

## Publications

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| <b>Human-Centered Program Synthesis</b><br>Ferdowsifard, K.                                                                                                                                    | Dissertation   |
| <b>Validating AI-Generated Code with Live Programming</b><br>Ferdowsi, K., Huang, R., James, M. B., Polikarpova, N., and Lerner, S.                                                            | CHI 2024       |
| <b>ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code</b><br>Ferdowsi, K., Williams, J., Drosos, I., Gordon, A.D., Negreanu, C., Polikarpova, N., Sarkar, A., and Zorn, B. | VL/HCC 2023    |
| <b>Investigating the Impact of Using a Live Programming Environment in a CS1 Course</b><br>Huang, R., Ferdowsi, K., Selvaraj, A., Soosai Raj, A.G., and Lerner, S.                             | SIGCSE TS 2022 |
| <b>LoopPy: Interactive Program Synthesis with Control Structures</b><br>Ferdowsifard, K., Barke, S., Peleg, H., Lerner, S., and Polikarpova, N.                                                | OOPSLA 2021    |
| <b>Small-Step Live Programming by Example</b><br>Ferdowsifard, K., Ordookhanians, A., Peleg, H., Lerner, S., and Polikarpova, N.                                                               | UIST 2020      |

## Research Experience

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| <b>Bottom-up Synthesis of Side-Effects with Separation Logic</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2024 |
| <ul style="list-style-type: none"><li>- With Hila Peleg.</li><li>- Reformulated bottom-up enumerative synthesis as proof search in Concrete Heap Separation Logic, allowing for efficient and provably correct synthesis in the presence of mutating components.</li><li>- Implemented and evaluated the technique, showing that it is comparable to the state-of-the-art while offering determinism and a correctness guarantee.</li><li>- Link to dissertation chapter: <a href="https://escholarship.org/uc/item/5cz057k7#page=108">https://escholarship.org/uc/item/5cz057k7#page=108</a></li></ul>                      |      |
| <b>LEAP: Validating AI-Generated Code with Live Programming</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2024 |
| <ul style="list-style-type: none"><li>- With Ruanqianqian (Lisa) Huang, Michael B. James, Sorin Lerner, and Nadia Polikarpova.</li><li>- Designed and implemented a novel interaction model for AI programming assistants that support exploring multiple code suggestions through Live Programming.</li><li>- Ran a 17 participant between-subjects user study showing numerous benefits to this interaction model, including preventing over-reliance, and reducing cognitive load.</li><li>- Link to publication: <a href="https://doi.org/10.1145/3613904.3642495">https://doi.org/10.1145/3613904.3642495</a></li></ul> |      |

## Research Experience (cont.)

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- Towards Human-Centered Types and Type Debugging** 2022
- PLATEAU'23 Workshop paper and presentation.
  - Surveyed the history of the usability of type systems, and type error messages. And offered concrete future directions based on cutting-edge HCI research.
  - Link to publication: <https://doi.org/10.1184/R1/22227457.v1>
- ColDeco: An End-User Spreadsheet Inspection Tool for AI-Generated Code** 2022
- With Jack Williams, Ian Drosos, Andrew Gordon, Carina Negreanu, Advait Sarkar, and Ben Zorn.
  - Designed and implemented a novel interaction model for inspecting and diagnosing errors in AI-generated code, targeted at end-users.
  - Conducted a 24-participant controlled experiment to evaluate the interaction.
  - Link to the paper: <https://ieeexplore.ieee.org/document/10305647>
- The Usability of Advanced Type Systems** 2022
- PhD Research Exam.
  - Surveyed publications on advanced type systems which inspired the Rust programming language, as well as all usability studies on Rust itself.
  - Link to the written report: <https://arxiv.org/abs/2301.02308>
- LoopPy: Interactive Program Synthesis with Control Structures** 2020 - 2021
- With Shraddha Barke, Hila Peleg, Sorin Lerner, and Nadia Polikarpova.
  - Designed and implemented a novel technique for using Bottom-up Enumerative Synthesis for efficient synthesis of blocks of Python code.
  - Links to the online editor, the paper, and conference talk: <http://loopy.goto.ucsd.edu/>
- Impact of Live Programming in a CS1 Course** 2020 - 2021
- With Ruanqianqian Huang, Ana Selvaraj, Adalbert Gerald Soosai Raj and Sorin Lerner.
  - Implemented a web editor for the Live Programming environment *Projection Boxes*, and developed and deployed a server for collecting usage data from the editor.
  - Ran a user study of hundreds of students in a CS1 course, and visualized and performed statistical analysis on data collected from the study.
  - Link to the paper: <https://dl.acm.org/doi/10.1145/3478431.3499305>
- SnipPy: Small-Step Live Programming by Example** 2020
- With Allen Ordookhanians, Hila Peleg, Sorin Lerner, and Nadia Polikarpova.
  - Designed and implemented a novel Program Synthesis interaction model with *Projection Boxes*.
  - Ran a 12-participant user study for evaluating the interaction.
  - Links to the online editor, the paper, and conference talk: <https://snippy.goto.ucsd.edu/>

## Engineering Experience

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- Senior Software Engineer** *Cubist Inc* 2024 - Present
- Building robust and secure software with Rust and TypeScript.
- Engineering Intern** *Tom Sawyer Software* Summer 2021
- Designed and developed the ability to deploy *Tom Sawyer Perspectives* applications to Kubernetes.
  - Lead the effort to document and improve *Tom Sawyer Perspectives'* deployment user experience.
- Engineering Intern** *Tom Sawyer Software* Summer 2020
- Co-designed and developed a tool using Git, Maven and Docker to streamline building and deploying *Tom Sawyer Perspectives* applications.
  - Main contributions included user-interface additions and improvements, an overhaul of the version control system, and bug fixes and improvements.

**Engineering Intern** *Tom Sawyer Software*

2016 - 2019

- Helped design and develop the *Tom Sawyer Licensing Server 2.0*, a backwards-compatible scalable web server using a microservice architecture.
- Implemented secure and user-friendly account management features, and a responsive web interface for *Graph Database Browser*, as well as database organization, query improvements, and other fixes.

**Teaching Experience**

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**Teaching Assistant** *CSE130: Programming Languages*

Fall 2023

- Presented discussion sections, held office hours, graded assignments, and assisted students on Piazza.

**Teaching Assistant** *CSE230: Graduate Programming Languages*

Fall 2022

- Presented discussion sections, held office hours, graded assignments, and assisted students on Piazza.

**Teaching Assistant** *CSE130: Programming Languages*

Winter 2022

- Presented discussion sections, held office hours, and assisted students on Piazza.

**Teaching Assistant** *CSE130: Programming Languages*

Fall 2021

- Presented discussion sections, held weekly office hours, and assisted students on Piazza.

**Teaching Assistant** *CSE11: Accelerated Intro to Programming*

Spring 2020

- Took responsibility for one week of the course in the instructor's absence, leading 2 lectures (in a live Q&A format with pre-recorded videos), and giving an online midterm exam to over 200 students.
- Led the weekly discussion sections, including writing review notes and practice problems tailored to the course, leading live discussions, and creating pre-recorded videos.

**Teaching Assistant** *CSE131: Compiler Construction*

Fall 2019

- Created autograder tests and infrastructure for the class projects.
- Assisted in the design of course assignments.
- Updated the compilation target from Intel x86 to x86\_64.

**Service**

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**Student Organizer**

PLATEAU 2024

**External Reviewer**

CHI 2023

**Program Committee Member**

HATRA 2023

**Student Volunteer**

PLDI 2023

**Mentorship**

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**GradWIC Mentor**

2022 - 2023

**CSE599: Teaching Methods in Computer Science Mentor**

Winter 2023

**ERSP Mentor**

2021 - 2022

- With Ruanqianqian (Lisa) Huang, Nadia Polikarpova and Sorin Lerner
- Link to poster: [https://drive.google.com/file/d/1HZx\\_FODLaYjbIDEKttTFO5jsF8EUov\\_Y/](https://drive.google.com/file/d/1HZx_FODLaYjbIDEKttTFO5jsF8EUov_Y/)